

flashscan

Installation guide — a read-only flash-loan arbitrage instrument

What it is

flashscan reads live decentralized-exchange prices across Ethereum, Base and Arbitrum, finds the best cross-DEX arbitrage route, and shows the real profit after every cost (flash-loan fee, gas, and price impact). It is read-only and honest: it never holds a private key, takes no deposit, and makes no income promise. On Ethereum mainnet you will mostly see “**no edge**” — that is the truth, not a bug. Anyone promising a crypto bot that “prints money” is selling you something.

Requirements

- git
- Python 3.10 or newer
- Optional: a web browser with the MetaMask extension, to connect a wallet
- Optional: a free RPC endpoint from alchemy.com (for live data)

Install in one command

Paste this into a terminal. It clones the repo, builds a Python virtualenv, installs dependencies, and runs the tests:

```
curl -fsSL https://raw.githubusercontent.com/dharaemon/chainsentinel/main/scripts/get-flashscan.sh | bash
```

Prefer to read the script first? The safer two-step is identical:

```
git clone https://github.com/dharaemon/chainsentinel.git
cd chainsentinel && bash scripts/install_flashscan.sh --no-serve
```

Launch the dashboard

```
cd chainsentinel
.venv/bin/python -m flashscan.server --port 8900
```

Then open **http://127.0.0.1:8900/** in your browser. For live data, add your own RPC key:

```
FLASHSCAN_RPC="https://eth-mainnet.g.alchemy.com/v2/YOUR_KEY" \
.venv/bin/python -m flashscan.server --port 8900
```

Using it

- **Demo** shows sample data instantly (offline). Switch **Source** to **Live** for real prices.
- **Chain**: Ethereum, Base, or Arbitrum. L2s (Base/Arbitrum) have thinner competition — your best shot at an edge.
- **Flash amount**: leave on **Auto**. Dragging it higher makes trades *worse* — a bigger loan moves the price against you.
- **Start monitoring**: re-scans every block and reports edge / no-edge live.

- **Execute** stays locked until a wallet is connected, a contract is set, and a real edge passes simulation. You sign every trade yourself.

Safety

- Read-only — it never sends a transaction on its own.
- **Never type your 12/24-word seed phrase into any website.** Ever.
- Use a throwaway wallet; practice on a testnet before mainnet.
- Not financial advice. No guaranteed profit.

Troubleshooting

- **“No pairs” on Live:** wrong RPC. Clear the RPC field or paste a real read RPC (Alchemy). Don't use the Flashbots relay as a read RPC.
- **RPC errors / 429:** use your own Alchemy key instead of public endpoints.
- **Base/Arbitrum empty:** set FLASHSCAN_RPC_BASE / FLASHSCAN_RPC_ARBITRUM to an L2 RPC.
- **Port busy:** pick another, e.g. --port 8901. Stop a running server with Ctrl+C.

Command-line quick reference

```
.venv/bin/python -m flashscan.cli --demo # dry-run, no network
.venv/bin/python -m flashscan.cli --chain base # live Base scan
.venv/bin/python -m flashscan.execution.cli plan --demo # build a route +
calldata
```

flashscan is an educational, defensive tool. The contract is written carefully but not professionally audited — read it, test on a testnet, and start tiny.